CIS437 - Cloud Computing

– Cloud Security I –
Foundations and Shared Responsibility

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Based on the original version by Professor Erik Fredericks

Last Class

- Regions and Zones
- Choosing Cloud Locations
- Latency and Availability
- Single-Region and Multi-Region Architecture
- Data Residency and Geographic Restrictions
- Geographic Access and Traffic Control
- Load Balancing

Practice



Scenario 1: An application has three VMs, but all three are in the same zone. Is it protected against a failure of that entire zone? Why or why not?

Scenario 2: A company must keep EU customer data in approved European locations. What factor should strongly influence its choice of cloud region?

Scenario 3: An application serves users in both the US and Europe, but all application servers are in the US. European users experience high latency. What architectural change could help?

Scenario 4: A company wants to deny requests originating from selected geographic regions. What type of cloud control could help implement this?

Scenario 5: An application runs in two regions and both regions can update customer data. What new data-management problem can arise?

Agenda

- Security Goals: CIA
- Threats and Common Cloud Security Risks
- Vulnerabilities and Patching
- Shared Responsibility
- Customer Security Responsibilities
- Defense in Depth
- Logging and Monitoring
- Practice

Opening Discussion: Security and Privacy

What are some security concerns we have?



And how about privacy?

Security Goals: The CIA Triad

Security Goals: CIA

Confidentiality

Only authorized users/services can access information.

Integrity

Data is not changed improperly or without authorization.

Availability

Systems and data remain accessible when needed.



If I put the data on an air-gapped drive and lock it away, is it secure?

Threats/Risks

Where do threats come from?

- People, bots, software/hardware failures, environmental events, third-party services

Internal or external?

- Both



Who?

- Humans, automated systems, technical/environmental causes

Why?

- **Malicious:** financial gain, data theft, disruption, curiosity/boredom
- **Non-malicious:** mistakes, misconfiguration, failures

AOHell

Threats/Risks

Threats can target:

- Confidentiality
- Integrity
- Availability

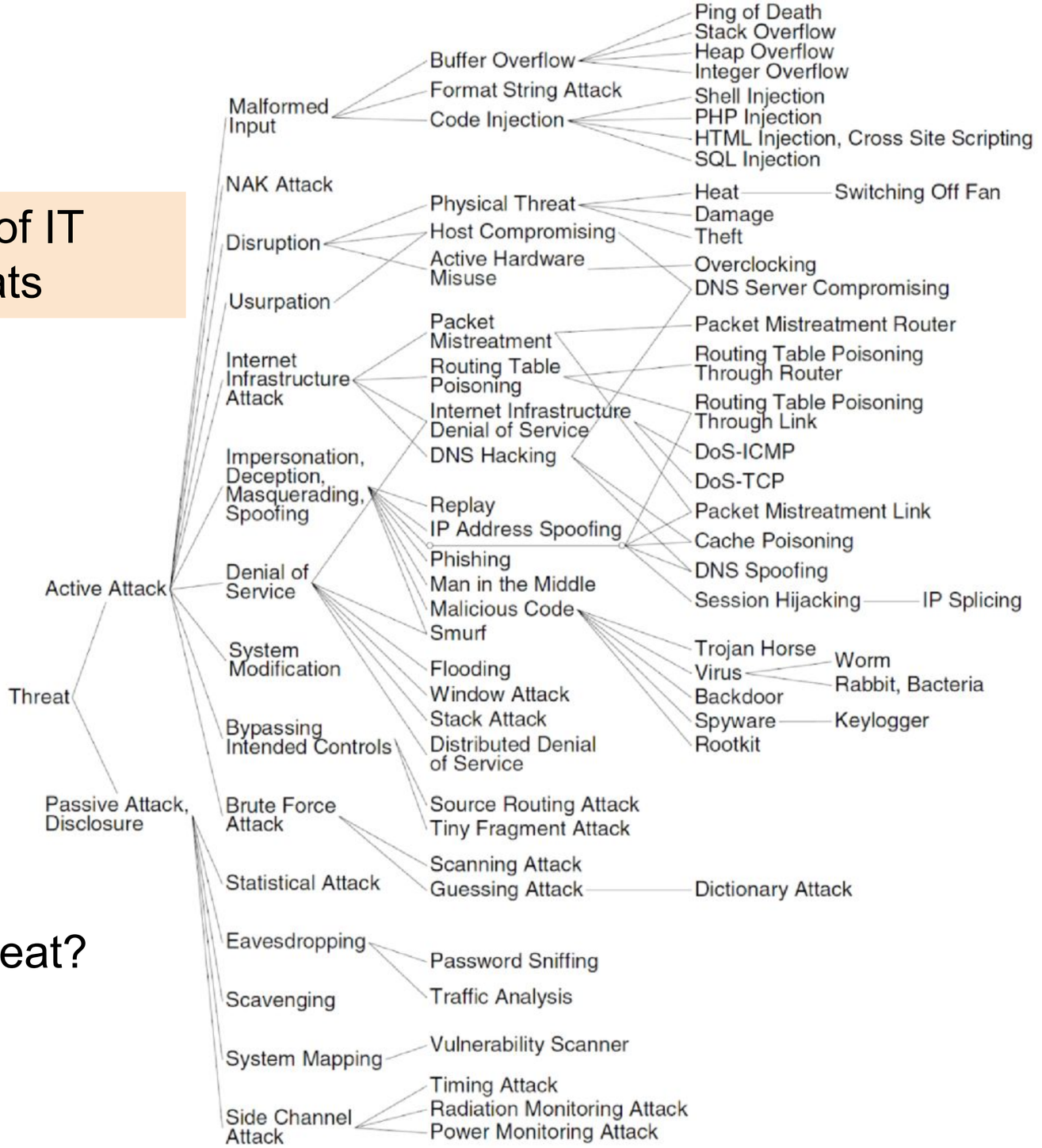
Threats can be:

- Physical
- Technical
- Human
- Environmental



Why is switching off a cooling fan a security threat?

Classification of IT Security Threats



Common Cloud Security Risks

- Misconfiguration
- Weak or stolen credentials
- Excessive permissions
- Publicly exposed resources
- Unpatched software
- Vulnerable or malicious dependencies
- Denial of service / resource abuse
- Insufficient logging and monitoring

Example: Software Supply Chain Risk

Compromised LiteLLM package steals developer and cloud credentials

Developer installs compromised dependency

↓

Package collects credentials/secrets

↓

Attacker may access cloud, source-code, or AI services



Vulnerabilities, Patching, and Staying Current

- Software vulnerabilities are continually discovered
- Internet-facing systems are automatically scanned
- Keep operating systems and software dependencies updated
- Remove or disable unnecessary services
- Monitor security advisories and vulnerabilities
- Managed services shift more infrastructure/runtime patching to the provider



If you manage a VM, keep it up to date.

Shared Responsibility: Provider vs. Customer

Cloud provider

- Physical facilities
- Hardware
- Core network/infrastructure
- Security of managed platform components

Customer

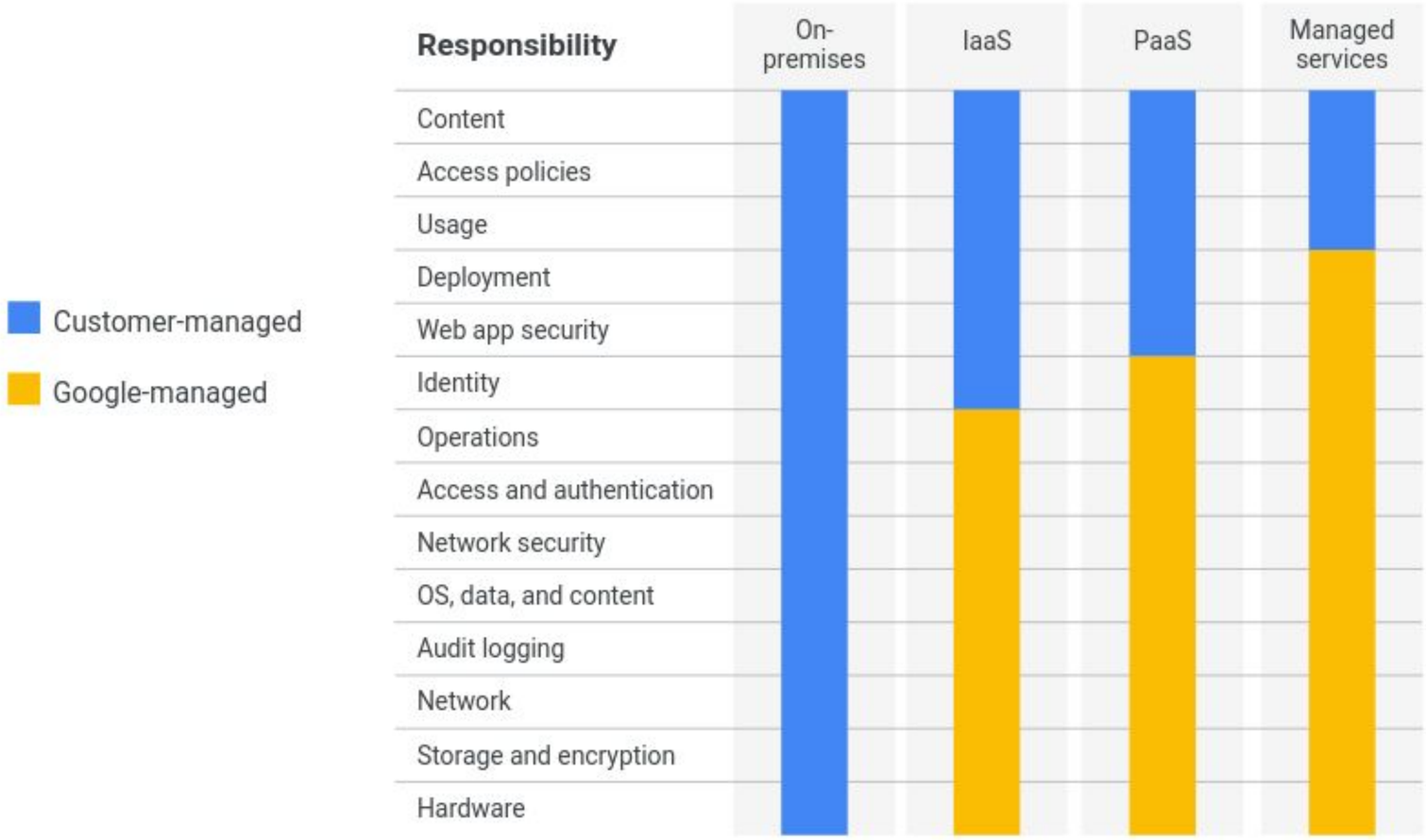
- Data
- Identities and permissions
- Application configuration
- Access controls
- Workload security



Who patches the operating system on a Compute Engine VM?

Your data + access decisions remain your responsibility

With Google Cloud, security responsibility is shared



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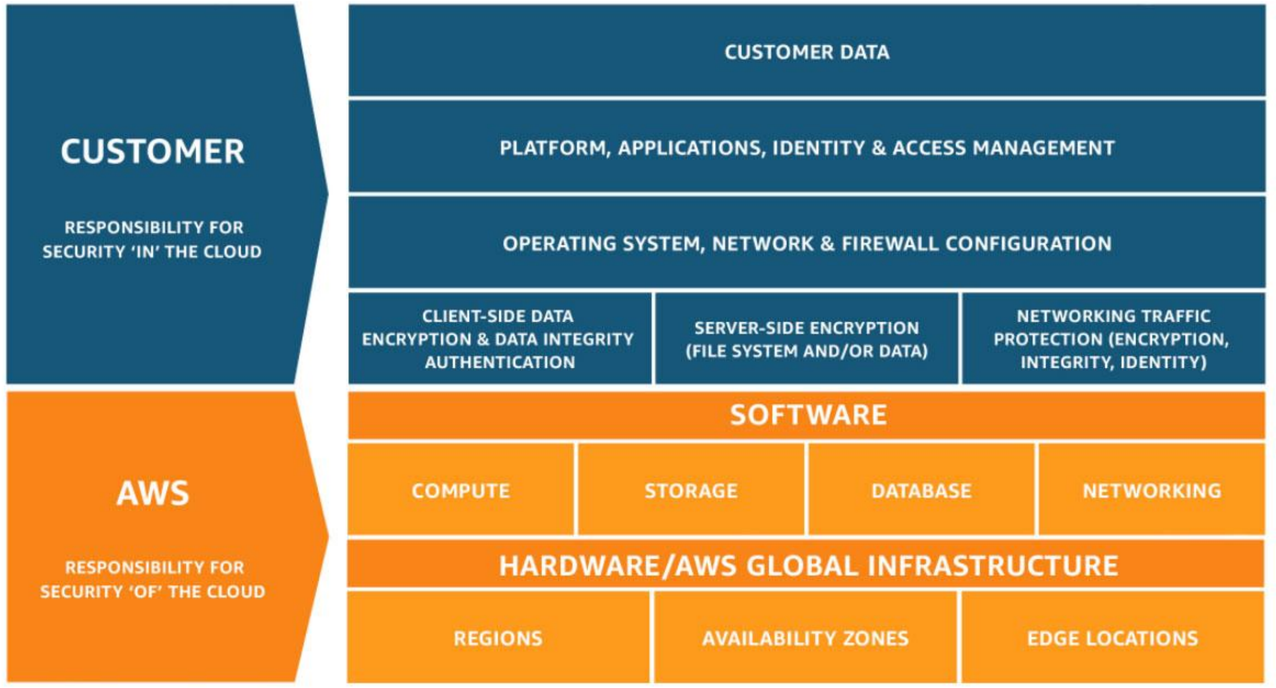
Responsibility scales

Compute Engine VM	Managed service / SaaS
More customer control	Less infrastructure control
Customer manages more configuration	Provider manages more infrastructure
Customer may manage OS and patches	Provider may manage runtime/platform
Customer still controls users/data/access	Customer still controls users/data/access

More managed service →

less infrastructure control →

less infrastructure responsibility, but users/data/access are still your responsibility.



Real Example: Weak Password on a Public VM

- Temporary Windows VMs used password Temp12345
- Automated attackers discovered some of the public VMs
- Compromised VMs were used for cryptocurrency mining
- Google detected the abuse and shut them down

Lesson: Cloud infrastructure security cannot compensate for weak customer configuration.

Your Security Responsibilities

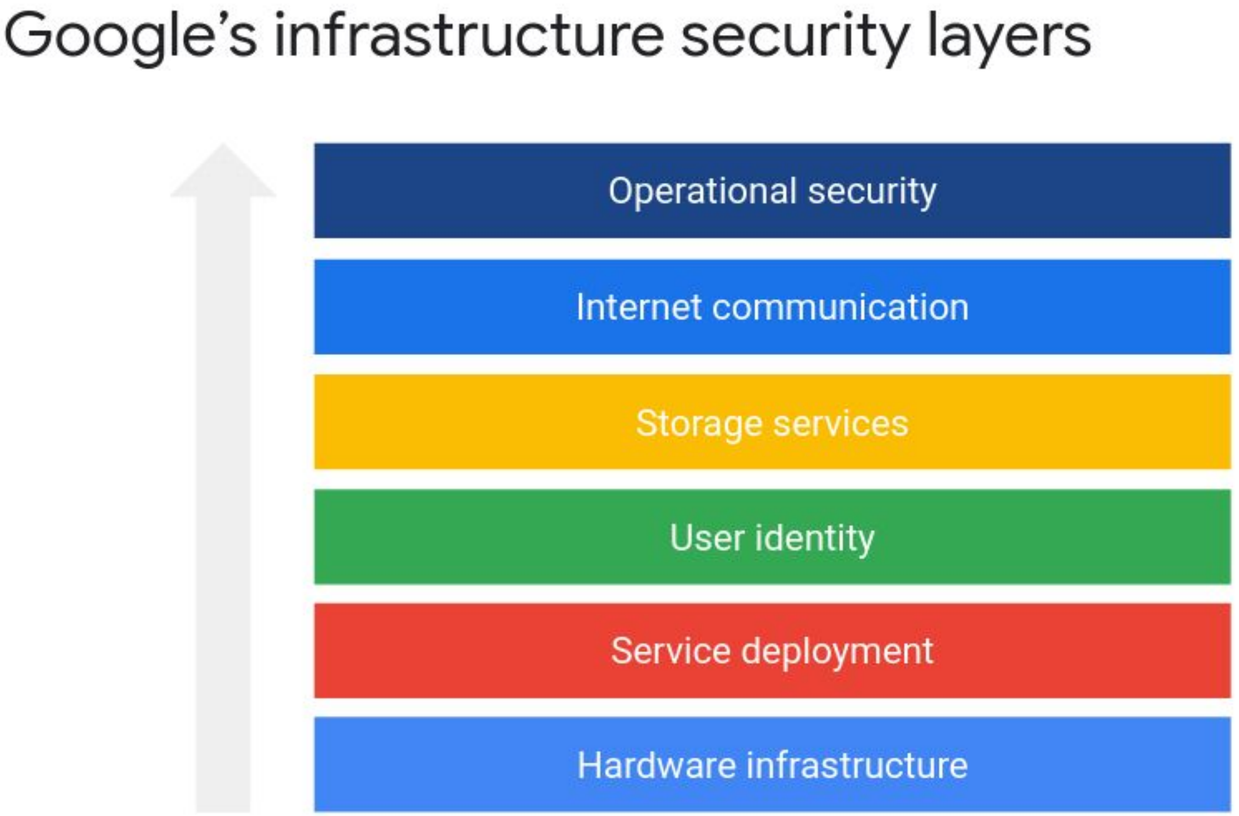
Depending on the service, customers may need to:

- Configure identities and permissions
- **Use strong authentication**
- **Apply least privilege**
- Configure network access and firewall rules
- Patch customer-managed software
- Protect credentials and secrets
- **Monitor access and system behavior**
- Respond to suspicious activity

The provider can build a strong wall, but if we leave the door open, the wall cannot protect us.

Defense in Depth

- Security should exist at multiple layers:
- Infrastructure
  Network
  Identity
  Application / service
  Data / storage
  Monitoring / operations



- If one control fails, another layer may still reduce the damage.

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Logging, Monitoring, and Alerts

- Collect logs from applications and cloud services
- Watch for errors and unusual activity
- Create alerts for important events
- Investigate unexpected access or usage

Example:

Cloud Run function fails repeatedly → log entry → alert → investigate → respond

Configure log-based alerting policies

Summary

- Confidentiality, integrity, and availability are core security goals
- Cloud threats can be malicious, accidental, technical, or environmental
- Misconfiguration, credentials, permissions, vulnerabilities, and dependencies are common risks
- Cloud security follows a shared-responsibility model
- Responsibility changes depending on how managed the cloud service is
- Customers must secure their workloads, identities, data, and configuration
- Defense in depth, logging, monitoring, and alerts reduce risk

Practice: Who Is Responsible: Provider or Customer?



- **Scenario 1:** A physical disk fails inside a Google data center.
  Who is responsible for the physical hardware?
- **Scenario 2:** A Compute Engine VM is configured with the password password123.
  Who is responsible for the weak authentication?
- **Scenario 3:** A developer accidentally makes a Cloud Storage bucket public.
  Who is responsible for the access configuration?
- **Scenario 4:** The physical network infrastructure supporting Cloud Storage has a hardware problem.
  Who is responsible for the underlying physical network?
- **Scenario 5:** A developer installs a malicious application dependency that steals a service credential.
  Who is responsible for managing this application/dependency risk?